

Construction of an Integrated Database for Protein Kinases and Proteases (IDPKP)

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ABSTRACT

Most protein kinases and protease families play important roles in cell communication and protein degradation in many biological pathways. Because of their importance, several useful databases such as the PKR and the MEROPS have been made. This paper focuses on kinases and proteases that are still hidden in the genome. Possible candidate proteins from human, mouse, and worm genome datasets were studied by using statistical models of conserved regions and sequence similarities. The non-redundant sets of screened proteins were combined with known proteins, resulting in the construction of a database called the Integrated Database of Protein Kinases and Proteases (IDPKP). The IDPKP is composed of 753 kinases and 971 proteases in the 'Known' category, and 1,746 kinases and 3,923 proteases in the 'Candidate' category.

SEARCH RESULT & STATISTICS

Protein Kinases

Species	Swiss-Prot Number of Protein	RefSeq (Local Link)	Source no Total	Only in Local Link	UnitGene Total	Whom Prep Total	GIS, Prod Total
Human	267	410	247	163	536	379	197
Mouse	187	236	144	92	467	316	233
Worm	43						

Proteases

Species	Swiss-Prot Number of Protein	RefSeq (Local Link)	Source no Total	Only in Local Link	UnitGene Total	Whom Prep Total	GIS, Prod Total
Human	289	571	247	334	1252	816	539
Mouse	183	280	156	132	1252	342	1369
Worm	47						

Summary

	Human	Mouse	Worm
Kinase			
Known Gene	431	279	43
Candidate	945	250	568
Protease			
Known Gene	607	317	47
Candidate	2270	1169	484

METHODS

PURPOSE

This project involves construction of an integrated DBMS for protein kinases and proteases of three distinctive species; *Homo sapiens*, *Mus musculus*, and *Caenorhabditis elegans*. Based upon the sources from which those proteins were extracted, they are largely grouped into 'Known' and 'Candidate' categories. The proteins in Known set may serve as templates for the future analysis and those in Candidate set, mostly derived from genome sequencing projects, may be the initial step in finding new protein kinases and proteases.

Mining Protein Kinases and Protease

To construct distinctive groups of each protein type, several families (groups) and motifs information had been used.

A. Grouping Sources

1. Protein Kinase

Source : Dr. Steven K. Hank' Classification[1]
Criterion : Similarity in catalytic domain
Groups : 5 major groups and 55 sub groups

2. Protease

Source : Pfam alignments[2] and MEROPS classification[3]
Criterion : Specificity and catalytic type
Groups : 6 major groups and 93 sub groups

Group Name	Description
AGC	Cyclic nucleotide related kinase
GMK	GTP/GAM related kinase
CKM	CDK/MAFK, GSK-3, Casein CK
PTX	Conventional protein tyrosine kinase
OPK	Other protein kinase

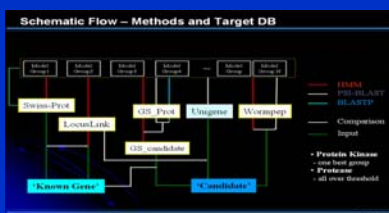
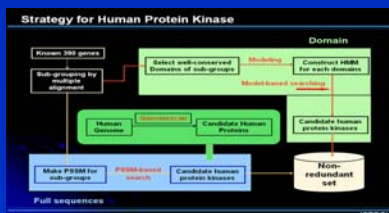
<Table 1. Protein Kinase Groups>

Group Name	Remark
Aspartate-type peptidases	Plan & MEROPS
Cysteine-type peptidases	Plan & MEROPS
Metalloproteinases	Plan & MEROPS
Serine-type peptidases	Plan & MEROPS
Peptidases of unknown catalytic mechanism	Plan & MEROPS
Other peptidases	Plan

<Table 2. Protease(peptidase) Groups>

Search Methods and Overall Schemes

- A. HMM : useful in finding specifically conserved domain.
- B. BLAST : overall similarity comparison

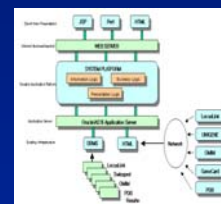


DBMS

General Information & Design

OS : Sun UNIX

DB : Oracle 9i Enterprise Edition



IMPLEMENTATION

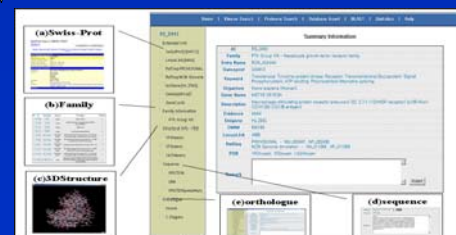
Availability : <http://www.istech.info/IDPKP/>

Protein Search and Results

Query type : Species, Family, ID/Accession, Keywords
Annotation (Known/Candidate), Disease



Detailed Protein Information



Other Features

Possible orthologue pairs with other species.
BLAST searches against the IDPKP sequences.

REFERENCES

- Hanks, S.K. and Quinn, A.M. (1991) Protein kinase catalytic domain sequence databases: Identification of conserved features of primary structure and classification of family members. *Methods Enzymol.* **200**, 38-62
- Bateman, A., Birney, E., et al. (2000) The Pfam protein families databases. *Nucleic Acids Res.* **28**, 263-266.
- Swiss-Prot Protein Knowledge Databases; <http://www.expasy.org/cgi-bin/list?peptides.txt>